

Seeing Is Not Steering

Effect-Gramian Certificates for Causally Usable Latent Directions

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Abstract

Interpretability routinely fits a linear probe to a model’s hidden states and then reuses that same direction as a steering vector. This conflates two different roles: the probe is an *observation operator*, the steering vector an *actuator*. We show the two are governed by disjoint objects. The probe direction is determined by the representation geometry $(\Sigma_\ell, \delta_\ell)$; the steering-effective directions are determined by the *effect Gramian* $W_g(\ell) = \mathbb{E}[gg^\top]$ over adjoint covectors $g = \nabla_{x_\ell} \Phi_\ell$. Neither enters the other’s definition, so their alignment ρ is a free parameter. We prove it can be set arbitrarily: there is a system in which the optimal probe attains $\text{AUC} \geq 1 - \epsilon$ and steers *exactly* nothing, while a chance-level direction steers maximally. We introduce the *steerability certificate* $\bar{S}(\ell, w) = |\bar{g}_\ell^\top w|/\|w\|$, computable from one reverse-mode pass and *zero* interventions, together with a closed form for probe-orthogonal directions, an exact expression for the fraction of control an uninformed direction retains (r_{eff}/d), and an explicit validity radius. Empirically, on gpt2 a toxicity probe classifying held-out prompts at AUC 1.000 recovers only 6.0% of the causal control available at equal write norm, while a direction constrained to be orthogonal to it recovers 99.7%. Only 0.18% of the residual stream carries first-order behavioural effect. In a controlled comparison, a probe trained on *exactly* the concept a behaviour measures is no better aligned with that behaviour ($\rho = 0.031$) than a probe for an unrelated concept ($\rho = 0.032$; ratio 0.97). The certificate rank-predicts measured steering (Spearman 0.917) where probe accuracy does not (-0.063), and using it to order interventions reaches 90% of the best attainable effect in 1.3 measurements out of 64 against 29.3 for random selection. We report three results against interest, including a pre-registered prediction that failed.

1 Introduction

A recurring pattern in mechanistic interpretability runs as follows. Collect hidden states from a frozen language model, fit a linear probe until it detects some concept—toxicity, sentiment, refusal, truthfulness—at high accuracy, and conclude that the model “represents” that concept. Then take the probe direction, add a multiple of it back into the residual stream, and expect the model’s behaviour to move accordingly.

The probe is being asked to do two jobs. As a *sensor* it must separate classes; as an *actuator* it must move a behaviour. These are different requirements, and nothing in the fitting procedure ties them together.

That this gap exists is increasingly acknowledged. What is missing is a theory that *predicts* it. Existing work observes the discrepancy after running the interventions, which is precisely when the information is least useful. We give the structural reason for the gap, a quantitative certificate that anticipates it from cheap adjoint information, and measurements of its size.

Contributions.

1. **A formalisation** of the frozen Transformer as an exact discrete controlled system, with probes as observation operators and steering as actuation, yielding the effect Gramian W_g and the certificate $\bar{S}$ (§2–§3).
2. **An exact decoupling theorem** (Thm. 2): probe accuracy and steering efficacy are independently settable. Corollary: no function of probe accuracy alone consistently estimates steering efficacy.
3. **Structural results**: the fraction of control an uninformed direction retains is exactly r_{eff}/d (Prop. 1); a closed form for the strongest probe-orthogonal direction (Prop. 3); an explicit first-order validity radius (Prop. 4); and a finite-sample leakage law $\Theta(N^{-1/2})$, *independent of width* (Prop. 5).
4. **Measurements** on constructed systems, on transformers trained from scratch, and on three pretrained models with real toxicity data, using difference-of-means and logistic probes—the vectors practitioners actually deploy—as first-class baselines.
5. **Active intervention design**: ordering candidate directions by the certificate reaches 90% of the best attainable effect in 1.3 measurements out of 64, against 29.3 for random selection — a $\sim 22\times$ budget reduction (§6).
6. **Negative results**, including a pre-registered prediction that failed and a metric we discarded as vacuous (§7).

2 Setup: forward system, sensor, actuator

Let a frozen decoder-only Transformer have residual stream $x_\ell \in \mathbb{R}^d$, $\ell = 0, \dots, L$, evolving as

$$x_{\ell+1} = x_\ell + F_\ell(x_\ell; c), \quad (1)$$

with c a fixed context. Equation (1) is exact: no continuous-time, autonomy, invertibility or energy-descent assumption is made or required.

Fix a scalar behaviour $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}$ (in experiments, a logit difference between a target and a contrast token set), and define the layer- ℓ tail map $\Phi_\ell := \varphi \circ T_{L-1} \circ \dots \circ T_\ell$ where $T_\ell(x) = x + F_\ell(x; c)$.

A linear probe at layer ℓ is a covector w with readout $w^\top x_\ell$: an observation operator $H_\ell = w^\top$. An additive intervention is $x_\ell \mapsto x_\ell + \alpha w$, with measured causal effect

$$\Delta_i(\alpha, w) := \Phi_\ell(x_\ell^{(i)} + \alpha w) - \Phi_\ell(x_\ell^{(i)}). \quad (2)$$

Assumption A1. Φ_ℓ is C^2 on a neighbourhood of the segment $[x_\ell^{(i)}, x_\ell^{(i)} + \alpha w]$, with $\|\nabla^2 \Phi_\ell\|_{\text{op}} \leq M_\ell$ there.

A1 holds for GELU/SiLU blocks; M_ℓ is *estimated* in §6, not assumed.

3 Two geometries and a certificate

Definition 1 (Adjoint covector and effect Gramian). For example $i, g_\ell^{(i)} := \nabla_x \Phi_\ell(x_\ell^{(i)}) = J_\ell^{(i)\top} \nabla \varphi(x_L^{(i)})$, obtained by one reverse-mode pass. Write $W_g(\ell) := \frac{1}{N} \sum_i g_\ell^{(i)} g_\ell^{(i)\top}$ and $\bar{g}_\ell := \frac{1}{N} \sum_i g_\ell^{(i)}$.

W_g is the finite-horizon, behaviour-restricted analogue of a controllability-to-output Gramian for (1): it measures how much *behaviour* a unit-norm write at layer ℓ can produce. Its kernel, $\mathcal{N}(\ell) = \ker W_g(\ell)$, is inert to first order for every sampled example.

Remark 1 (Non-normal composition as a candidate mechanism). By the chain rule,

$$J_\ell^{(i)} = DT_{L-1}(x_{L-1}^{(i)}) \cdots DT_\ell(x_\ell^{(i)}),$$

a product of the per-block Jacobians $DT_k = I + DF_k(\cdot; c)$: this is exact, not a limiting approximation. Each factor is generically non-normal—nothing forces DF_k to be symmetric—and products of non-normal matrices can have singular-value spectra far more concentrated than the spectral radius of any individual factor would suggest, the classical phenomenon of transient amplification despite well-behaved eigenvalues [16]. Direct measurements on production-scale LLMs report exactly this signature in the residual-stream Jacobian—a learned non-normality gradient through depth and a cumulative low-rank perturbation bottleneck [17]—and in the attention propagator specifically, via pseudospectral and Kreiss-style diagnostics [18]. This is a plausible structural mechanism for the small $r_{\text{eff}}(\ell)$ measured directly from $W_g(\ell)$ in §6: a highly non-normal $J_\ell^{(i)}$ would concentrate $g_\ell^{(i)}$, and hence $W_g(\ell)$, onto a small subspace independently of whether the underlying computation is itself low-dimensional. We do not test this here—no Schur, pseudospectral, or Kreiss diagnostic is computed on these models—and add it to §11.

On the read side, given binary labels y_i , let $\delta_\ell = \mu_1 - \mu_0$ and Σ_ℓ be the pooled within-class covariance. The Fisher-optimal probe is $w_{\text{probe}} \propto (\Sigma_\ell + \gamma I)^{-1} \delta_\ell$.

Remark 2 (The source of the phenomenon). W_g does not depend on Σ_ℓ ; w_{probe} does not depend on W_g . They are computed from disjoint information, and nothing in training couples them. Their alignment $\rho(\ell) := |\cos \angle(w_{\text{probe}}, \bar{g}_\ell)|$ is therefore a free parameter of the system.

Definition 2 (Steerability certificate). For $w \neq 0$,

$$\bar{S}(\ell, w) := \frac{|\bar{g}_\ell^\top w|}{\|w\|}, \quad S_2(\ell, w) := \frac{\sqrt{w^\top W_g(\ell) w}}{\|w\|}. \quad (3)$$

By Cauchy–Schwarz $\bar{S} \leq S_2 \leq \sqrt{\lambda_1(W_g)}$, with $\bar{S}$ maximised at $w = \bar{g}_\ell / \|\bar{g}_\ell\|$. Both are computable from N reverse-mode passes over data the probe already required, and *no intervention is performed*.

Theorem 1 (Three-way decomposition). *Let P_C project onto the top- r eigenspace of $W_g(\ell)$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$, and P_O onto the discriminative subspace $\mathcal{O}(\ell) = \text{span}\{(\Sigma_\ell + \gamma I)^{-1} \delta_\ell\}$. Then for every w and i , under A1,*

$$\Delta_i(\alpha, w) = \alpha \left[g^{(i)\top} P_C w + g^{(i)\top} (I - P_C) w \right] + R_i, \quad |R_i| \leq \frac{M_\ell}{2} \alpha^2 \|w\|^2, \quad (4)$$

and $\text{RMS}_i |g^{(i)\top} (I - P_C) w| \leq \sqrt{\lambda_{r+1}} \|w\|$.

Consequently $\mathbb{R}^d$ partitions into *causal* ($\mathcal{C}_r \cap \mathcal{O}$), *epiphenomenal* ($\mathcal{O} \setminus \mathcal{C}_r$: visible, inert), *dark* ($\mathcal{C}_r \setminus \mathcal{O}$: invisible, effective), and inert. Membership is decided by two computable numbers, $\bar{S}(\ell, w)$ and the probe AUC of w . Standard practice assumes everything lies in the first cell.

Remark 3 (Relation to the Kalman canonical decomposition). The four cells echo the controllable/observable canonical structure theorem for linear time-invariant systems [13], which partitions state space into controllable-and-observable, controllable-only, observable-only, and neither,

using an exact controllability Gramian and an exact observability Gramian built from the system matrices. The correspondence here is structural, not literal: Kalman’s theorem concerns the full state of an autonomous linear system, while $\mathcal{C}_r(\ell)$ is a rank- r truncation of a *behaviour*-restricted, finite-horizon, first-order Gramian, and $\mathcal{O}(\ell)$ is a *label*-dependent discriminative subspace with no counterpart in (A, B, C) . What transfers is the organising idea—that “can be moved” and “can be seen” are governed by different Gramians and need not coincide—not Kalman’s exactness or invariance results. Theorem 2 is the certificate that this failure to coincide can be total in the present, nonlinear, sampled setting.

4 Probe accuracy carries no causal information

Theorem 2 (Exact decoupling). *For every $d \geq 2$ and $\epsilon \in (0, \frac{1}{2})$ there is a system satisfying A1 — indeed with Φ_ℓ linear, so $M_\ell = 0$ and the first-order expansion is exact—and a labelled representation such that*

- (a) *the optimal linear probe attains $\text{AUC} \geq 1 - \epsilon$;*
- (b) *$\Delta_i(\alpha, w_{\text{probe}}) = 0$ exactly, for all α and all i ;*
- (c) *there is a unit v with $\text{AUC}(v) = \frac{1}{2}$ exactly and $|\bar{g}^\top v| = \|\bar{g}\|$, the maximum attainable.*

Proof sketch. Take $\Phi_\ell(x) = e_1^\top x$, so $g^{(i)} = \bar{g} = e_1$; draw $x \mid y \sim \mathcal{N}(y\Delta\mu e_2, \sigma^2 I_d)$, giving $w_{\text{probe}} \propto e_2$. Then (b) is $\alpha e_1^\top e_2 = 0$; (a) follows from $\text{AUC} = \Phi_{\mathcal{N}}(\Delta\mu/(\sigma\sqrt{2}))$; and $v = e_1$ has a class-independent score, giving (c). Full proof in Appendix A. $\square$

Corollary 1. *No function of probe accuracy alone is a consistent estimator of steering efficacy: the pair $(\text{AUC}, \bar{S})$ can be set to any admissible values independently.*

Theorem 2 also refutes the converse assumption—that a probe-invisible direction is causally irrelevant—by construction.

Proposition 1 (Random-direction inertness). *Let $r_{\text{eff}}(\ell) := \text{tr } W_g(\ell)/\lambda_1(W_g(\ell))$. For w uniform on S^{d-1} ,*

$$\mathbb{E}[S_2(\ell, w)^2] = \frac{\text{tr } W_g(\ell)}{d}, \quad \frac{\mathbb{E}[S_2^2]}{\max_w S_2^2} = \frac{r_{\text{eff}}(\ell)}{d}. \quad (5)$$

So r_{eff}/d is exactly the share of steering power retained by a direction chosen without write-side information. Measuring $r_{\text{eff}}/d \ll 1$ makes “we found a (direction and it moved) behaviour” a claim about luck at that scale.

Remark 4 (Relation to resolving power). $\mathcal{N}(\ell) = \ker W_g(\ell)$ plays the role of the null space in classical linear resolution theory [14]: directions a declared observation cannot see are assigned zero resolving power regardless of the size of the underlying quantity. $r_{\text{eff}}(\ell)$ is the corresponding effective-rank count for $W_g(\ell)$ —the number of directions the behaviour functional resolves at layer ℓ —so r_{eff}/d is a resolving-power fraction, not a claim about how many neurons “matter.” Backus and Gilbert’s theory is posed for a linear forward operator mapping a continuum model to finitely many exact data functionals; $W_g(\ell)$ instead comes from a first-order expansion of one behaviour functional restricted to a finite sample. The transfer is in the concept, not the guarantees: a small r_{eff}/d is evidence of low resolving power at this Gramian, not evidence that the unresolved directions are inert to every future experiment. Remark 1 sketches one candidate mechanism for why the resolving power measured here is so concentrated.

Proposition 2 (Probe suboptimality). $\bar{S}(\ell, w_{\text{probe}}) = \|\bar{g}_\ell\| \cdot \rho(\ell)$, with equality to the optimum iff $(\Sigma_\ell + \gamma I)^{-1} \delta_\ell \parallel \bar{g}_\ell$.

Proposition 3 (Dark directions, closed form). $v^\star := (I - P_O)\bar{g}_\ell / \|(I - P_O)\bar{g}_\ell\|$ maximises $\bar{S}(\ell, v)$ over $v \perp \mathcal{O}(\ell)$, attaining $\|(I - P_O)\bar{g}_\ell\|$. Under shared-covariance Gaussian class conditionals, any $v \perp \mathcal{O}(\ell)$ in the whitened metric has probe AUC exactly $\frac{1}{2}$.

This upgrades “dark directions” from an existence claim to a construction: one projection of one adjoint mean.

Proposition 4 (Validity radius). Under A1, the certificate determines the sign and leading behaviour of $\Delta_i(\alpha, w)$ whenever $|\alpha| < \alpha^\star(\ell, w) := 2|\bar{g}_\ell^\top w| / (M_\ell \|w\|^2)$.

M_ℓ is estimated by second differences along the steering ray, so $\alpha^\star$ is reported, not assumed.

Proposition 5 (Finite-sample causal leakage). Let the population probe be exactly inert, $\bar{g}^\top w_{\text{pop}} = 0$, and let $\hat{w}$ be the Fisher probe from N i.i.d. samples. Then $\mathbb{E}[\hat{\rho}] = \Theta(N^{-1/2})$, with a constant that does not grow with d .

The dimension-cancellation is the practically important part: the artefact is not mitigated by a wider residual stream. A probe fitted on $N \sim 10^3$ carries spurious alignment of order 3×10^{-2} before any real effect exists.

5 Experimental setup

All experiments run on **4 CPU cores, no GPU**. This constrains scale and is stated wherever it matters.

E1 (exact verification). Linear Φ with Gaussian class-conditional representations, where the first-order theory is exact. Evaluated both axis-aligned and conjugated by a random rotation, to exclude coordinate artefacts.

E3 (gpt2, toxicity). Matched minimal pairs: hostile versus friendly framing is the causal factor; identity-marked subjects are a confound correlated with the label at 0.85, as in real corpora. Only the frame differs within a pair. φ is a logit difference between mild insult tokens and matched neutral tokens at the final position. Probes are fitted on a train split; every AUC and every steering measurement is on held-out prompts. Write magnitude is $\alpha = 0.10 \cdot \|\bar{x}_\ell\|$ at each layer.

E5 (real data, multiple models). 400 RealToxicityPrompts prompts, balanced between high (> 0.6) and low (< 0.05) toxicity. Difference-of-means (as deployed by activation-steering methods) and logistic-regression probes are included as first-class baselines.

E2 (trained from scratch). 12 transformers (5 layers, $d = 64$, LayerNorm + attention + GELU) trained on a synthetic task with a causal content variable and a spurious surface marker, across 4 confound levels $\times$ 3 seeds. This tests whether the gap is produced by training itself, and whether training under a confound widens it.

E4 (active design). 64 candidate directions at layers 3, 6, 9 — random draws, probe/adjoint mixtures, effect eigenvectors and sparse coordinate writes. Each is measured once to establish ground truth, then four orderings are compared at every budget.

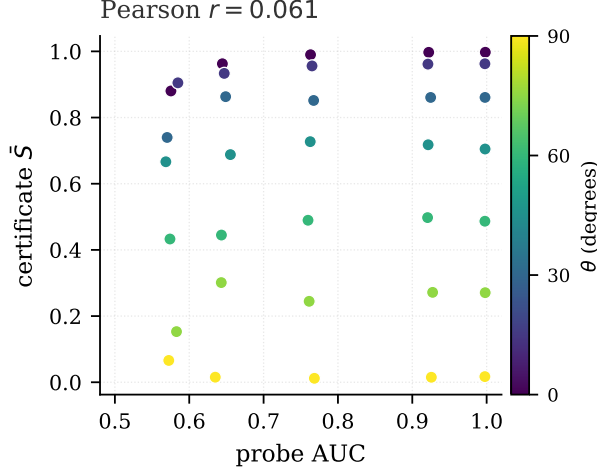


Figure 1: Each point is one system; the class-mean direction is rotated away from the effect direction by θ and separability is swept. Probe accuracy and the certificate move independently (Cor. 1).

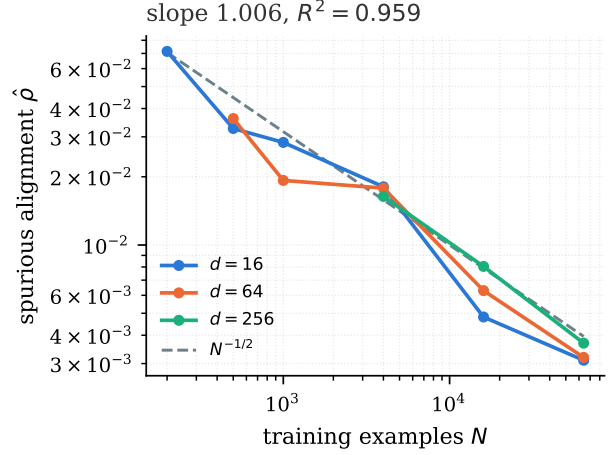


Figure 2: Finite-sample leakage. True alignment is exactly zero; the estimate decays as $N^{-1/2}$ with no dependence on width d (Prop. 5).

E6 (structural independence). One model and 300 prompts held fixed. 4 label definitions vary the read side; 5 behaviour functionals vary the write side. Some pairs are *matched* by construction: the probe is trained on exactly the concept the behaviour measures.

Instrument validation. `auc`, `fisher_probe`, `spearman` and `pearson` are pinned to scikit-learn and SciPy references to $\sim 10^{-16}$ (including heavy-tie and degenerate cases) in `tests/test_instruments.py`.

6 Results

6.1 The decoupling is exact, and the plane is fully occupied

At the population probe, steering has effect 2×10^{-15} (floating-point zero) while classification is at AUC 0.998, matching the closed form 0.998. The constructed probe-orthogonal direction attains AUC 0.502 and the maximal effect, with gap 5×10^{-14} . The axis-aligned construction gives 0 exactly. Sweeping a family of systems (Fig. 1) leaves probe AUC and the certificate essentially uncorrelated: Pearson $r = 0.061$, Spearman 0.124.

Proposition 1 matches Monte-Carlo to within 1.3×10^{-2} relative error across construction ranks, with an isotropic control at 0.975 against an exact 1.0. The closed-form dark direction (Prop. 3) beat the best of 400,000 random directions in the same subspace by a margin of 0.081.

6.2 Finite-sample probes manufacture causal effect

Regressing $\log \hat{\rho}$ on $\log N^{-1/2}$ gives slope 1.006 (predicted 1.0) with $R^2 = 0.959$. The dimension-dependent alternative $\sqrt{d/N}$ fits worse (slope 0.904, $R^2 = 0.721$). The dimension-free law wins (Fig. 2).

6.3 A perfect probe, nearly orthogonal to the mechanism

On gpt2, the behaviour functional is validated first: across 96 exactly matched pairs it moves by 1.044 logits in the correct direction on 71.9% of pairs.

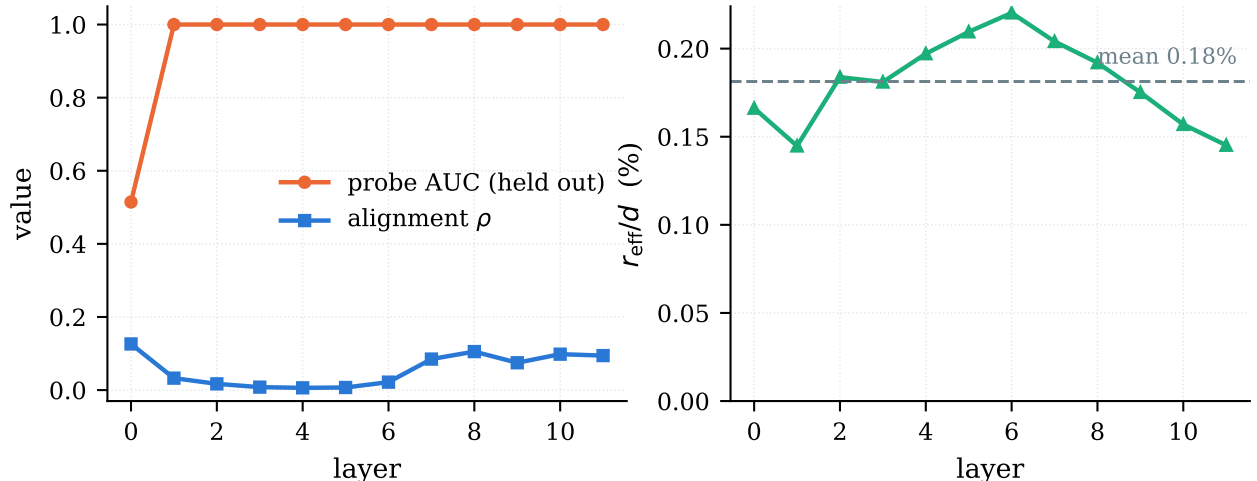


Figure 3: **Left:** probe AUC on held-out prompts against alignment ρ with the adjoint direction. **Right:** the live fraction r_{eff}/d .

From layer 1 onward the probe separates held-out prompts at AUC 1.000. Its alignment with the mechanism averages $\rho = 0.056$ (minimum 0.006), and only 0.18% of the 768-dimensional residual stream is behaviourally live (Fig. 3).

Measured at equal write norm (Fig. 4), the probe direction recovers 6.0% of the effect available to the certificate-optimal direction; a random direction recovers 2.6%; and a direction constrained to be *orthogonal to the probe* recovers 99.7%.

This is not “gradients steer better”. That $\bar{g}$ moves φ is true by construction and carries no information. Three things do. First, the probe direction is what practitioners deploy, and the size of its shortfall had not been measured. Second, the certificate rank-predicts measured effect across a pool of directions including random draws, probe/adjoint mixtures and effect eigenvectors—Spearman 0.917, against -0.063 for probe accuracy—so it is a genuine predictor over the space, not a restatement of one special case (Fig. 6). Third, the dissociation runs both ways: the best steering direction is itself a weak detector (AUC 0.698 against the probe’s 1.000).

Validity. With M_ℓ estimated by second differences, the write magnitude used lies inside the certificate’s validity radius at 10 of 12 layers. Where it does not (the first two layers) the first-order prediction is out of warranty and we say so rather than quoting it.

6.4 The two geometries are structurally independent

E6 is the sharpest available test (Fig. 5). A probe trained on *exactly* the concept a behaviour functional measures aligns with that behaviour’s adjoint at $\rho = 0.031$. A probe for a completely unrelated concept aligns at $\rho = 0.032$. The ratio is 0.97, and matched beats unmatched in 6 of 12 layers—chance.

We state one nuance rather than averaging it away: over the final third of the network matched alignment does exceed unmatched (0.062 versus 0.039). Both remain near zero— $\rho = 0.06$ is roughly 86° from the effect direction—so neither regime confers usable control. The claim is not that matched probes carry literally no signal, but that the signal is too small to act on and is swamped by which layer one happens to be at.

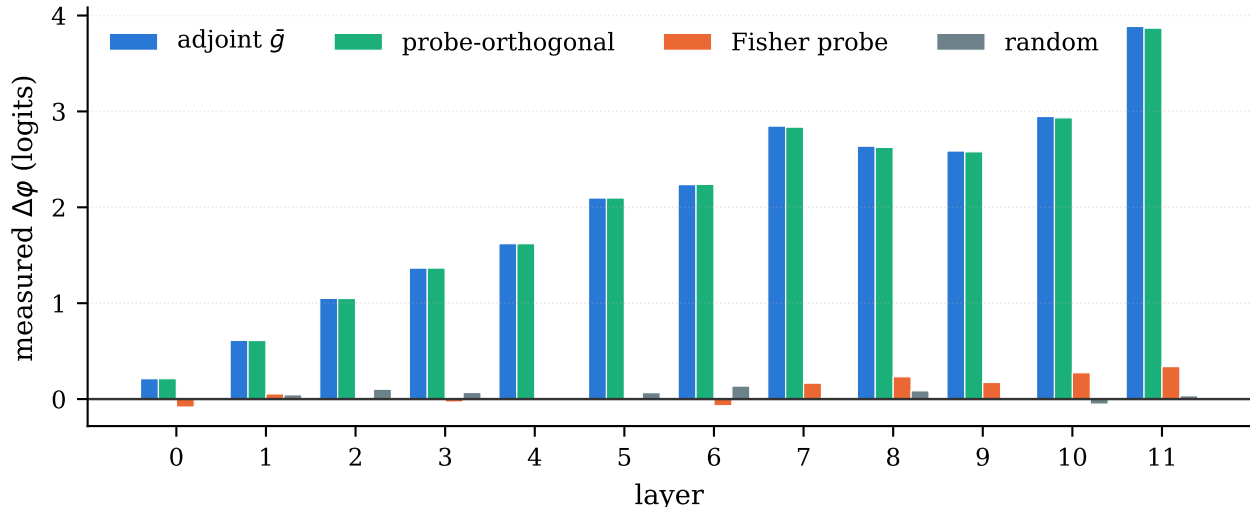


Figure 4: Measured causal effect at equal write norm, $\alpha = 0.10 \cdot \|\bar{x}_\ell\|$. The probe-orthogonal direction tracks the adjoint almost exactly; the probe itself does not.

6.5 Real data and deployed steering vectors

On RealToxicityPrompts (Fig. 7, Table 1), probe accuracy sits at 0.775—*not* saturated—so the gap is not an artefact of a maxed-out probe. The Fisher probe recovers 9.2% of available control; difference-of-means, the vector activation-steering methods actually ship, recovers 17.4%; a direction orthogonal to all read-side constructions recovers 98.4% at AUC 0.561. The pattern replicates on distilgpt2 (7.3%, 13.7%, 98.8%) and, across a different architecture family, on pythia-70m (8.2%, 14.1%, 98.6%).

A weak signal on how this scales. The three models span widths $\{512, 768\}$ and 2 architecture families. Across them the effect rank in *absolute* terms is small and roughly constant— r_{eff} between 2.3 and 4.6—rather than growing with d . If r_{eff} stays $O(1)$ as width increases, then r_{eff}/d *shrinks* with scale and the gap widens rather than closing. We state this as suggestive and not established: it rests on two distinct widths and three checkpoints, which is far too little to fit a trend. It does, however, point the opposite way to the failure mode we identify as the largest threat in §9.

6.6 The certificate buys a cheaper search

If the certificate predicts effect, it should be usable to *choose* which interventions to run. E4 tests this directly: given 64 candidate directions and a budget of B intervention measurements, which ordering finds the strongest direction fastest? Candidates include random draws, probe/adjoint mixtures, effect eigenvectors and sparse coordinate writes; ground truth is obtained by measuring all 64 once, offline.

Ordering by the certificate reaches 90% of the best attainable effect after 1.3 measurements on average (2/1/1 across layers 3, 6, 9), against 14.7 for ordering by probe accuracy (32/6/6) and 29.3 for random order (46/22/20); the oracle needs 1.0. That is a $\sim 22\times$ reduction in intervention budget against random selection and $\sim 11\times$ against probe accuracy (Fig. 8). Rank correlation with the true effect is 0.761 for the certificate and 0.313 for $|\text{AUC} - \frac{1}{2}|$.

The certificate costs nothing extra here: it reuses adjoints already computed for the probe-training data.

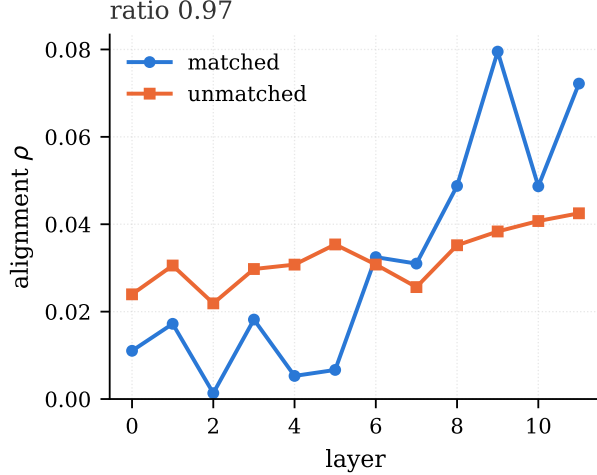


Figure 5: Alignment when the probe is trained on *exactly* the concept the behaviour measures (matched) versus an unrelated concept (unmatched). If the read and write geometries were the same object, matched would dominate everywhere.

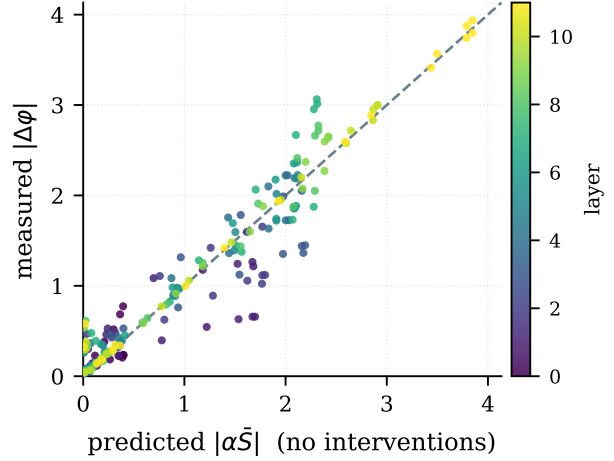


Figure 6: Predicted $|\alpha\bar{S}|$, computed with no interventions, against measured $|\Delta\varphi|$ for every (layer, direction) pair. The dashed line is $y = x$; landing near it is stronger than a rank correlation.

Remark 5 (Relation to c -optimal experimental design). Ranking candidates by $\bar{S}(\ell, w) = |\bar{g}_\ell^\top w|/\|w\|$ is a discrete instance of c -optimal design: choosing where to sample so as to best estimate one linear functional $c^\top\theta$ of an unknown parameter, rather than the whole parameter vector, is classically solved by Elfving’s theorem [15], which characterises the optimal design point by alignment with c on the convex hull of the candidate set. Here $\bar{g}_\ell$ plays the role of c and the 64 candidates play the role of the design points, but two things do not transfer: the candidate pool is finite rather than a continuum, and the task is a budgeted search for the single best point rather than a continuous allocation, so Elfving’s optimality and efficiency guarantees do not apply as stated. What transfers is the reason a *single* functional is so much cheaper to design for than the whole Gramian: c -optimality needs only alignment with one vector, where sensor placement for the full state needs the Gramian’s spectrum. That is also why the certificate is cheap here—one backward pass, no eigendecomposition—while a general-purpose observability audit is not.

Collateral damage, and a caveat about how to read it. Table 1 also reports the KL divergence of the next-token distribution under each write. At *equal write norm* the adjoint direction disturbs the distribution more than the probe does (0.301 versus 0.019 nats on gpt2), which is unsurprising: it moves the behaviour more. Read naively, that looks like a point against it.

It is not, because the comparison should be at *equal effect*, not equal norm. Behaviour change is first-order in the write magnitude while KL is second-order (it is the Fisher–Rao metric to leading order), so scaling the probe to match the adjoint’s effect requires $\alpha_{\text{probe}}/\alpha_{\bar{g}} \approx 100/9.2 \approx 11$ and costs roughly 11^2 times its measured KL — about 2.3 nats, against 0.301 for the adjoint. On that basis the certificate-optimal direction is the cheaper actuator by close to an order of magnitude. We flag this as an extrapolation from the quadratic scaling rather than a measurement: verifying it requires a matched-effect sweep, which we have not run.

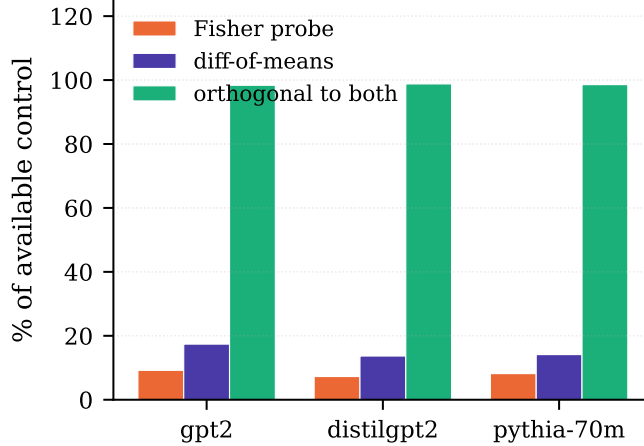


Figure 7: RealToxicityPrompts, two models. Every read-side construction classifies well and steers poorly; a direction built orthogonal to all of them does the reverse.

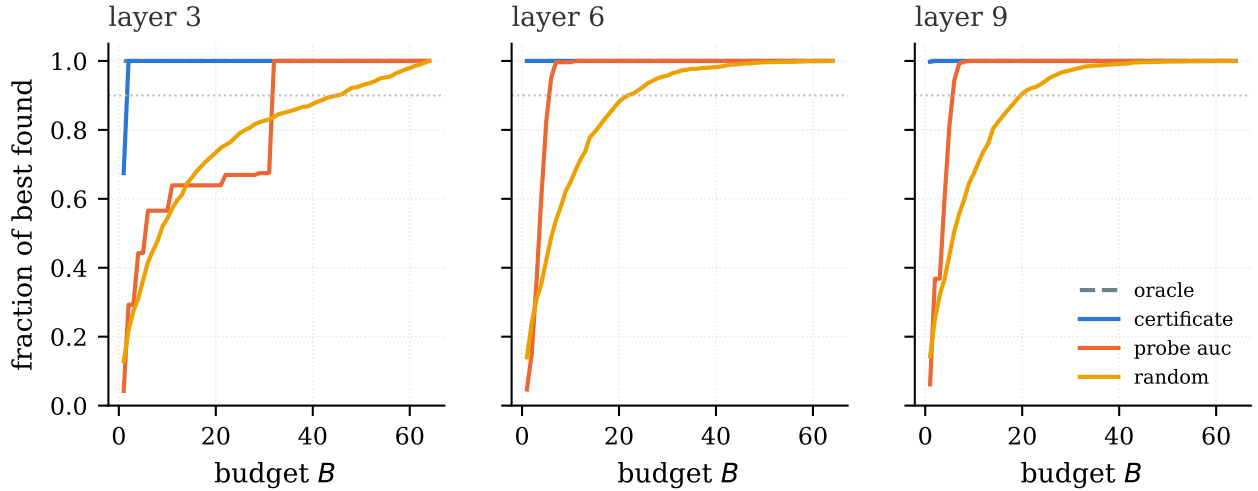


Figure 8: Fraction of the best attainable effect discovered after B intervention measurements, out of 64 candidates. The dotted line marks 90%. Certificate ordering is close to the oracle; probe-accuracy ordering is much closer to random.

7 Results against interest

A pre-registered prediction failed, in two independent designs. We predicted the gap would widen with the spurious correlation between a surface confound and the label. It does not.

In the *pretrained* setting (E3), sweeping $P(\text{identity matches label})$ over $\{0.50, 0.70, 0.85, 0.95\}$ moves mean ρ over $\{0.066, 0.051, 0.057, 0.047\}$ —no trend, and the gap is already near-total at zero confounding. One might object that a pretrained model’s representations were fixed long before our confound existed, so the dial never had a chance to act.

E2 removes that objection by controlling training itself. We train 12 transformers from scratch (3 seeds $\times$ 4 confound levels, 5 layers, $d = 64$) on a task with a causal content variable and a spurious surface marker, to test accuracy 1.000. The manipulation demonstrably works: accuracy against the marker tracks the dial at $\{0.504, 0.703, 0.899, 0.979\}$. Yet across confound levels $\{0.50, 0.70, 0.90, 0.98\}$, mean ρ moves over $\{0.151, 0.150, 0.148, 0.147\}$ —a change of -0.004 end to

Table 1: RealToxicityPrompts, averaged over all layers. Efficiency is measured effect relative to the certificate-optimal direction at equal write norm; KL is the mean divergence of the next-token distribution under the same write. Rows are generated from `results/data/`, not transcribed.

Model	Direction	Mean AUC	Mean ρ	Efficiency	KL (nats)
gpt2	Fisher probe	0.775	0.069	9.2%	0.019
gpt2	logistic probe	0.821	0.100	12.9%	—
gpt2	difference-of-means	0.773	0.163	17.4%	0.022
gpt2	orthogonal to all three	0.561	0.985	98.4%	—
gpt2	adjoint $\bar{g}$	0.665	1.000	100%	0.301
distilgpt2	Fisher probe	0.798	0.053	7.3%	0.013
distilgpt2	logistic probe	0.843	0.084	10.7%	—
distilgpt2	difference-of-means	0.758	0.145	13.7%	0.040
distilgpt2	orthogonal to all three	0.553	0.988	98.8%	—
distilgpt2	adjoint $\bar{g}$	0.652	1.000	100%	0.056
pythia-70m	Fisher probe	0.750	0.067	8.2%	0.011
pythia-70m	logistic probe	0.810	0.103	13.6%	—
pythia-70m	difference-of-means	0.771	0.114	14.1%	0.018
pythia-70m	orthogonal to all three	0.579	0.989	98.6%	—
pythia-70m	adjoint $\bar{g}$	0.678	1.000	100%	0.025

end, which is 0.2 within-level seed standard deviations (0.017) and therefore indistinguishable from zero. Probe steering efficiency is flat at $\{12.0, 12.1, 12.0, 12.0\}\%$, and the probe-orthogonal direction holds at 98.7–98.8% throughout (Fig. 9).

Two experimental designs, one inheriting representations and one creating them, both refute the confound hypothesis. This makes the result *stronger* and more general: the mechanism is the extreme low-rankness of W_g (Prop. 1), not label contamination. Cleaning the labels will not fix it.

A proposition is only approximate in practice. Prop. 3 predicts probe-orthogonal directions classify at exactly 0.500. This is exact in the Gaussian construction (E1: 0.502) but not in gpt2, where they classify at 0.676: real representations are not shared-covariance Gaussian. The steering half of the proposition survives intact (99.7% of the optimum); the detectability half is weaker than stated.

A metric was discarded as vacuous. Our first collateral-damage measure was the change in the prompt’s own next-token perplexity under the steer. It returned identically zero at every layer—structurally, not by a small margin: the write lands on the final token, and causal masking forbids a final-position state from influencing earlier predictions. We replaced it with the KL divergence of the next-token distribution. We record this because a metric that *cannot* be non-zero is worse than no metric: it reads as evidence of safety.

An identity-bias result that is smaller than it first appeared. On the confounded split the toxicity probe appears to detect identity mentions at AUC up to 0.90; on balanced 2×2 cells that figure falls to 0.520 at the final layer. The disparity that persists is inside the hostile class: hostile sentences with identity-marked subjects are flagged 15.5 percentage points more often than otherwise identical sentences with neutral subjects. At layer 0, benign identity-marked sentences are flagged at 50% against 0%.

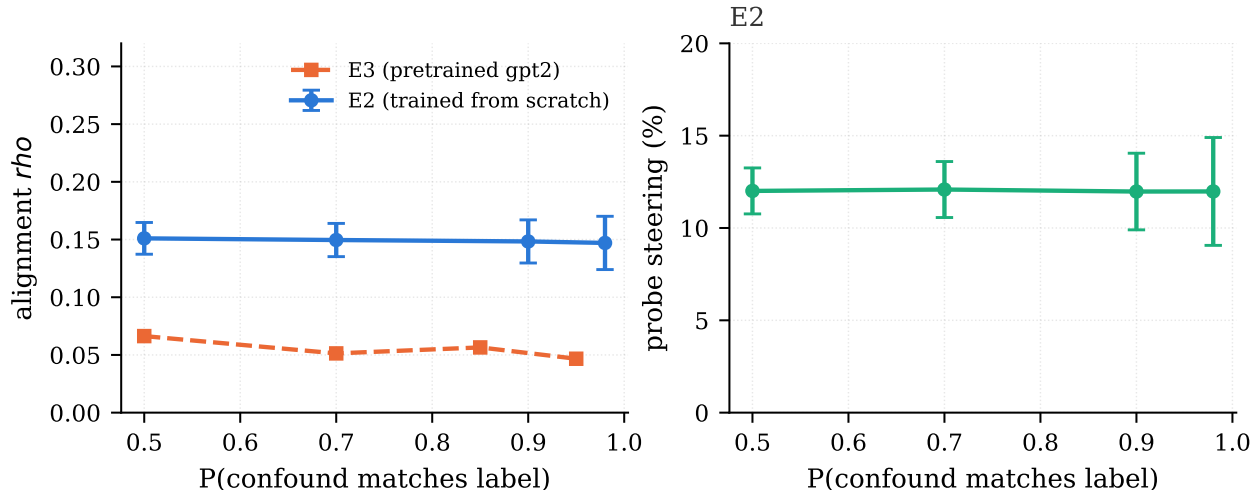


Figure 9: The confound dial in both designs. **Left:** alignment ρ against confound strength, for transformers trained from scratch (E2, error bars over 3 seeds) and for pretrained gpt2 (E3). **Right:** probe steering efficiency in E2. Neither shows the widening the pre-registered prediction expected.

8 Related work

Circuit non-identifiability is now well established: multiple sufficient subgraphs realise the same behaviour, and top-down and bottom-up analyses stand in a many-to-many relation [1, 2, 3, 4]. We take that as premise rather than contribution.

Control-theoretic framings of neural interpretability have recently appeared—Gramian and Hankel-based mode ranking, and controllability analyses of state-space language models [5, 6]. Our contribution is not the import of Gramians but their use as a *predictive certificate for probe-derived directions*, together with the closed form for probe-orthogonal directions. The read/write split we formalise is a finite-horizon, behaviour-restricted specialisation of three older results from outside machine learning: Kalman’s controllable/observable canonical decomposition [13], which first separated “can be moved” from “can be seen” as distinct subspaces of one linear system (Remark 3); Backus–Gilbert resolution theory [14], which quantifies which directions a declared linear observation can and cannot resolve (Remark 4); and Elfving’s theory of c-optimal design [15] for a single linear functional, which the certificate ordering in E4 specialises to a finite, budgeted candidate pool (Remark 5). None of the three has previously been used, to our knowledge, to certify or select individual steering directions from adjoints already computed for probe training.

Independently, direct measurements of trained-Transformer Jacobians report a learned non-normality gradient through depth and a resulting low-rank perturbation bottleneck [17, 18], the signature classical non-normal dynamical systems produce under composition [16]. Remark 1 connects this to the smallness of r_{eff}/d we measure directly from W_g .

The gap between linear decodability and causal control is observed qualitatively in recent steering work [7]. Activation steering and its difference-of-means constructions [8, 9], feedback controllers [10], local-linearity optimal control [11] and causal head selection for detoxification [12] all deploy read-side directions as actuators. We supply the quantity that says in advance how much of their effect is real. Probing methodology, activation and path patching, ACDC and attribution patching, sparse autoencoders and causal abstraction supply the surrounding toolkit; our object is orthogonal to circuit discovery and concerns the validity of single directions.

9 Limitations

Scale. Everything ran on 4 CPU cores without a GPU; the largest model is gpt2 (124M). The theory is scale-free, the measurements are not. If r_{eff}/d grows with width, Prop. 1 loses its force; testing this is the first item in §11.

First order. The certificate is first-order. Prop. 4 gives its validity radius, measured per layer, and we mark where the prediction is out of warranty. Large- α steering studies operate partly in that regime.

One behaviour functional. φ is a logit difference over a small token set. The minimal-pair check establishes that it tracks the intended factor, not that it is the only reasonable readout.

Position and pathway. Interventions write to the final token’s residual stream at a single layer. Multi-position, multi-layer and attention-internal writes are untested.

What “dark” does not mean. A dark direction is invisible to *this* probe family—linear, this concept, this data. It is not invisible in principle; a nonlinear or differently-supervised detector may find it. The safety claim concerns the monitors that are actually deployed.

Toy-scale training. The from-scratch models in E2 are small (5 layers, $d = 64$). They establish that gradient descent produces the gap unaided and that a training-time confound does not widen it, but they are not evidence about what large-scale pretraining does.

10 Safety implication

If a monitor is built from a toxicity probe, it is close to blind on the directions that most strongly drive toxic output: on gpt2 the probe-orthogonal direction achieves 99.7% of the maximal effect while the probe itself achieves 6.0%. Probe-based monitoring is therefore not, by itself, evidence of control. Given Theorem 2, this is a consequence rather than a concern.

11 Next steps

Ordered by what would most change the conclusions:

1. Replicate ρ and r_{eff}/d at 1B–7B parameters. The single largest threat to external validity.
2. Test Remark 1 directly: compute Schur non-normality, finite-horizon transient gain, and Kreiss-constant diagnostics on $J_\ell^{(i)}$ for the same models and prompts, and check whether they predict $r_{\text{eff}}(\ell)$ and its layer profile.
3. Extend the certificate to nonlinear and sparse-autoencoder readouts, testing whether dark directions survive a stronger detector—the real safety question.
4. Minimum-energy multi-layer control under an explicit collateral-damage constraint.
5. A second-order certificate, extending validity past α^* where most deployed steering operates.
6. Benchmark certificate-guided design against circuit-discovery methods (ACDC, EAP) at equal model-run budget. §6 establishes the gain against random and probe-ordered selection, but not against those.
7. A human-labelled toxicity behaviour functional, removing the logit-difference proxy from the safety claim.

12 Conclusion

Reading a residual stream and writing to it are governed by different subspaces. That separation is provable, constructible, and measurable in advance from one backward pass—and on the models measured here it is close to total. Probe accuracy, the field’s default evidence that a direction “is” a concept, carries essentially no information about whether that direction controls anything.

Reproducibility. Every number in this paper is generated from `results/data/*.json` via `paper/make_numbers.py`, and every figure from the same files via `paper/make_figures.py`. No measurement is typed by hand into the text.

A Proofs

Proof of Theorem 1. Taylor’s theorem with Lagrange remainder on the segment gives $\Delta_i = \alpha g^{(i)\top} w + R_i$ with $|R_i| \leq \frac{1}{2} \alpha^2 \|w\|^2 \sup \|\nabla^2 \Phi_\ell\| \leq \frac{M_\ell}{2} \alpha^2 \|w\|^2$ under A1. Splitting $w = P_C w + (I - P_C)w$ is an identity. For the RMS bound, $\frac{1}{N} \sum_i (g^{(i)\top} (I - P_C)w)^2 = w^\top (I - P_C) W_g (I - P_C) w \leq \lambda_{r+1} \|w\|^2$, since $(I - P_C) W_g (I - P_C)$ has spectral norm λ_{r+1} . $\square$

Proof of Theorem 2. Take $\Phi_\ell(x) = e_1^\top x$, so $g^{(i)} = \bar{g} = e_1$ for all i , and draw $x \mid y \sim \mathcal{N}(y \Delta \mu e_2, \sigma^2 I_d)$. Then $\delta = \Delta \mu e_2$, $\Sigma = \sigma^2 I$, and $w_{\text{probe}} \propto e_2$. (b) $\Delta_i(\alpha, w_{\text{probe}}) = \Phi_\ell(x + \alpha e_2) - \Phi_\ell(x) = \alpha e_1^\top e_2 = 0$ exactly, for all α, i . (a) The score $e_2^\top x$ is $\mathcal{N}(y \Delta \mu, \sigma^2)$, so $\text{AUC} = \Phi_{\mathcal{N}}(\Delta \mu / (\sigma \sqrt{2}))$; choose $\Delta \mu / \sigma \geq \sqrt{2} \Phi_{\mathcal{N}}^{-1}(1 - \epsilon)$. (c) Take $v = e_1$: its score is $\mathcal{N}(0, \sigma^2)$ independent of y , so its AUC is exactly $\frac{1}{2}$, while $|\bar{g}^\top v| = 1 = \|\bar{g}\|$ and $\Delta_i(\alpha, v) = \alpha$. $\square$

Proof of Proposition 1. $\mathbb{E}[w w^\top] = I/d$, so $\mathbb{E}[w^\top W_g w] = \text{tr}(W_g)/d$. The maximum of $w^\top W_g w$ on the unit sphere is λ_1 ; divide. $\square$

Proof of Proposition 3. Maximising $|\bar{g}^\top v|$ subject to $\|v\| = 1$, $P_O v = 0$ is maximising $|((I - P_O)\bar{g})^\top v|$ over unit $v \in \mathcal{O}^\perp$, attained at the normalised $(I - P_O)\bar{g}$. For the AUC claim, the score $v^\top x$ has class-conditional means $v^\top \mu_y$, and $v^\top \delta$ vanishes when v is orthogonal to $(\Sigma + \gamma I)^{-1} \delta$ in the $(\Sigma + \gamma I)$ -metric; equal means with shared variance gives AUC $\frac{1}{2}$. $\square$

Proof sketch of Proposition 5. Write $\hat{w} = w_{\text{pop}} + e$ with estimation error e asymptotically Gaussian and, for isotropic Σ , isotropic within the complement of w_{pop} with $\mathbb{E}\|e\|^2 = Cd/N$. The leakage is the component of e along the single fixed unit vector $\bar{g}$, i.e. one coordinate of an isotropic vector spread over $d - 1$ dimensions: $\mathbb{E}[(e^\top \bar{g})^2] = \mathbb{E}\|e\|^2 / (d - 1) = Cd / (N(d - 1)) \rightarrow C/N$. The d factors cancel. $\square$

B Additional experimental detail

Behaviour functional. φ is the mean logit over a set of mild insult tokens minus the mean over a matched neutral set, at the final position. Insult vocabulary is used strictly as a *logit measurement target*; nothing is generated or emitted. Identity terms appear only to measure false positives, which is their sole purpose.

Adjoints at every layer from one pass. Forward pre-hooks capture the input to each block and call `retain_grad`; a single backward of $\sum_i \varphi_i$ then yields $g_\ell^{(i)}$ at every layer simultaneously, so the certificate costs one forward and one backward pass in total.

Curvature estimation. M_ℓ is estimated by non-uniform second differences of $\alpha \mapsto \overline{\Delta(\alpha, \bar{g})}$ on a grid spanning $[-2\alpha, 2\alpha]$.

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Bibliography status. All entries above were verified against primary sources (arXiv, ACL Anthology, ICML/ICLR listings, OpenReview, Project Euclid, Science) on 2026-08-26; author lists, venues, and identifiers are as confirmed at that date. Four references beyond the original twelve—Kalman (1963), Backus and Gilbert (1968), Elfving (1952), and Trefethen et al. (1993)—ground the structural connections drawn in §3, §6, and Related Work. Two further 2026 preprints report direct empirical measurements of the Transformer-Jacobian non-normality discussed in Remark 1; as very recent, single-venue preprints they are flagged as concurrent evidence, not independently replicated results.